CHAPTER 1

PATTERNS IN MATHEMATICS

CBSE / NCERT Curriculum • Standard Grade 6 Mathematics

CORE TOPICS & LEARNING OBJECTIVES:

1. Chapter Introduction & Conceptual Overview

In this chapter, students investigate the foundational principles of Patterns in Mathematics. Mathematics is not merely about mechanical computations; it is about discovering relationships, finding symmetries, and applying structured reasoning to solve real-world problems.

NCERT Ganita Prakash introduces these concepts through tactile explorations, visual models, and guided inquiries. Students are encouraged to observe patterns carefully before formulating generalized rules.

KEY EXAMPLE 1.1 (WORKED DEMONSTRATION):

Problem Statement: Find the 10th triangular number: $T_n = n(n+1)/2 = 10(11)/2 = 55$.

Step-by-step Solution: Follow standard NCERT methodology with full justifications.

NCERT DIDACTIC EXPLORATION CORNER:

Remember: Always verify your calculation by checking special cases and boundary values.

1.2 Triangular Numbers and Geometric Arrangements

Mathematical analysis for Patterns in Mathematics continues on page 2. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw the 5th dot pattern of square numbers and calculate 5 x 5 = 25.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.2 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.5]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.6]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.7]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.8]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.3 Square Numbers and Dot Grids

Mathematical analysis for Patterns in Mathematics continues on page 3. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the 10th triangular number: $T_n = n(n+1)/2 = 10(11)/2 = 55$.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.3 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.9]

Given the conditions of 1.3 Square Numbers and Dot Grids, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.10]

Given the conditions of 1.3 Square Numbers and Dot Grids, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.11]

Given the conditions of 1.3 Square Numbers and Dot Grids, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.12]

Given the conditions of 1.3 Square Numbers and Dot Grids, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.4 Number Sequences and Arithmetic Differences

Mathematical analysis for Patterns in Mathematics continues on page 4. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Explain the rule for the sequence: 4, 7, 10, 13, 16... (Rule: Add 3, n-th term = $3n + 1$).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.13]

Given the conditions of 1.4 Number Sequences and Arithmetic Differences, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.14]

Given the conditions of 1.4 Number Sequences and Arithmetic Differences, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.15]

Given the conditions of 1.4 Number Sequences and Arithmetic Differences, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.16]

Given the conditions of 1.4 Number Sequences and Arithmetic Differences, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.5 The Fibonacci Sequence in Nature

Mathematical analysis for Patterns in Mathematics continues on page 5. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw the 5th dot pattern of square numbers and calculate 5 x 5 = 25.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.17]

Given the conditions of 1.5 The Fibonacci Sequence in Nature, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.18]

Given the conditions of 1.5 The Fibonacci Sequence in Nature, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.19]

Given the conditions of 1.5 The Fibonacci Sequence in Nature, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.20]

Given the conditions of 1.5 The Fibonacci Sequence in Nature, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.1 Visual Patterns and Sequences

Mathematical analysis for Patterns in Mathematics continues on page 6. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the 10th triangular number: $T_n = n(n+1)/2 = 10(11)/2 = 55$.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.21]

Given the conditions of 1.1 Visual Patterns and Sequences, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.22]

Given the conditions of 1.1 Visual Patterns and Sequences, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.23]

Given the conditions of 1.1 Visual Patterns and Sequences, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.24]

Given the conditions of 1.1 Visual Patterns and Sequences, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.2 Triangular Numbers and Geometric Arrangements

Mathematical analysis for Patterns in Mathematics continues on page 7. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Explain the rule for the sequence: 4, 7, 10, 13, 16... (Rule: Add 3, n-th term = $3n + 1$).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.25]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.26]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.27]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.28]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.3 Square Numbers and Dot Grids

Mathematical analysis for Patterns in Mathematics continues on page 8. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw the 5th dot pattern of square numbers and calculate 5 x 5 = 25.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.29]

Given the conditions of 1.3 Square Numbers and Dot Grids, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.30]

Given the conditions of 1.3 Square Numbers and Dot Grids, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.31]

Given the conditions of 1.3 Square Numbers and Dot Grids, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.32]

Given the conditions of 1.3 Square Numbers and Dot Grids, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.4 Number Sequences and Arithmetic Differences

Mathematical analysis for Patterns in Mathematics continues on page 9. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the 10th triangular number: $T_n = n(n+1)/2 = 10(11)/2 = 55$.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.33]

Given the conditions of 1.4 Number Sequences and Arithmetic Differences, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.34]

Given the conditions of 1.4 Number Sequences and Arithmetic Differences, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.35]

Given the conditions of 1.4 Number Sequences and Arithmetic Differences, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.36]

Given the conditions of 1.4 Number Sequences and Arithmetic Differences, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.5 The Fibonacci Sequence in Nature

Mathematical analysis for Patterns in Mathematics continues on page 10. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Explain the rule for the sequence: 4, 7, 10, 13, 16... (Rule: Add 3, n-th term = 3n + 1).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.37]

Given the conditions of 1.5 The Fibonacci Sequence in Nature, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.38]

Given the conditions of 1.5 The Fibonacci Sequence in Nature, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.39]

Given the conditions of 1.5 The Fibonacci Sequence in Nature, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.40]

Given the conditions of 1.5 The Fibonacci Sequence in Nature, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.1 Visual Patterns and Sequences

Mathematical analysis for Patterns in Mathematics continues on page 11. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw the 5th dot pattern of square numbers and calculate $5 \times 5 = 25$.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.41]

Given the conditions of 1.1 Visual Patterns and Sequences, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.42]

Given the conditions of 1.1 Visual Patterns and Sequences, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.43]

Given the conditions of 1.1 Visual Patterns and Sequences, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.44]

Given the conditions of 1.1 Visual Patterns and Sequences, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.2 Triangular Numbers and Geometric Arrangements

Mathematical analysis for Patterns in Mathematics continues on page 12. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the 10th triangular number: $T_n = n(n+1)/2 = 10(11)/2 = 55$.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.45]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.46]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.47]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.48]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.3 Square Numbers and Dot Grids

Mathematical analysis for Patterns in Mathematics continues on page 13. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Explain the rule for the sequence: 4, 7, 10, 13, 16... (Rule: Add 3, n-th term = $3n + 1$).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.49]

Given the conditions of 1.3 Square Numbers and Dot Grids, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.50]

Given the conditions of 1.3 Square Numbers and Dot Grids, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.51]

Given the conditions of 1.3 Square Numbers and Dot Grids, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.52]

Given the conditions of 1.3 Square Numbers and Dot Grids, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.4 Number Sequences and Arithmetic Differences

Mathematical analysis for Patterns in Mathematics continues on page 14. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw the 5th dot pattern of square numbers and calculate 5 x 5 = 25.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.53]

Given the conditions of 1.4 Number Sequences and Arithmetic Differences, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.54]

Given the conditions of 1.4 Number Sequences and Arithmetic Differences, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.55]

Given the conditions of 1.4 Number Sequences and Arithmetic Differences, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.56]

Given the conditions of 1.4 Number Sequences and Arithmetic Differences, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.5 The Fibonacci Sequence in Nature

Mathematical analysis for Patterns in Mathematics continues on page 15. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the 10th triangular number: $T_n = n(n+1)/2 = 10(11)/2 = 55$.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.57]

Given the conditions of 1.5 The Fibonacci Sequence in Nature, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.58]

Given the conditions of 1.5 The Fibonacci Sequence in Nature, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.59]

Given the conditions of 1.5 The Fibonacci Sequence in Nature, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.60]

Given the conditions of 1.5 The Fibonacci Sequence in Nature, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.1 Visual Patterns and Sequences

Mathematical analysis for Patterns in Mathematics continues on page 16. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Explain the rule for the sequence: 4, 7, 10, 13, 16... (Rule: Add 3, n-th term = $3n + 1$).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.61]

Given the conditions of 1.1 Visual Patterns and Sequences, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.62]

Given the conditions of 1.1 Visual Patterns and Sequences, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.63]

Given the conditions of 1.1 Visual Patterns and Sequences, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.64]

Given the conditions of 1.1 Visual Patterns and Sequences, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.2 Triangular Numbers and Geometric Arrangements

Mathematical analysis for Patterns in Mathematics continues on page 17. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw the 5th dot pattern of square numbers and calculate $5 \times 5 = 25$.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.65]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.66]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.67]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.68]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.3 Square Numbers and Dot Grids

Mathematical analysis for Patterns in Mathematics continues on page 18. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the 10th triangular number: $T_n = n(n+1)/2 = 10(11)/2 = 55$.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.69]

Given the conditions of 1.3 Square Numbers and Dot Grids, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.70]

Given the conditions of 1.3 Square Numbers and Dot Grids, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.71]

Given the conditions of 1.3 Square Numbers and Dot Grids, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.72]

Given the conditions of 1.3 Square Numbers and Dot Grids, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.4 Number Sequences and Arithmetic Differences

Mathematical analysis for Patterns in Mathematics continues on page 19. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Explain the rule for the sequence: 4, 7, 10, 13, 16... (Rule: Add 3, n-th term = $3n + 1$).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.73]

Given the conditions of 1.4 Number Sequences and Arithmetic Differences, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.74]

Given the conditions of 1.4 Number Sequences and Arithmetic Differences, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.75]

Given the conditions of 1.4 Number Sequences and Arithmetic Differences, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.76]

Given the conditions of 1.4 Number Sequences and Arithmetic Differences, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.5 The Fibonacci Sequence in Nature

Mathematical analysis for Patterns in Mathematics continues on page 20. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw the 5th dot pattern of square numbers and calculate 5 x 5 = 25.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.77]

Given the conditions of 1.5 The Fibonacci Sequence in Nature, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.78]

Given the conditions of 1.5 The Fibonacci Sequence in Nature, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.79]

Given the conditions of 1.5 The Fibonacci Sequence in Nature, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.80]

Given the conditions of 1.5 The Fibonacci Sequence in Nature, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.1 Visual Patterns and Sequences

Mathematical analysis for Patterns in Mathematics continues on page 21. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the 10th triangular number: $T_n = n(n+1)/2 = 10(11)/2 = 55$.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.81]

Given the conditions of 1.1 Visual Patterns and Sequences, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.82]

Given the conditions of 1.1 Visual Patterns and Sequences, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.83]

Given the conditions of 1.1 Visual Patterns and Sequences, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.84]

Given the conditions of 1.1 Visual Patterns and Sequences, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

1.2 Triangular Numbers and Geometric Arrangements

Mathematical analysis for Patterns in Mathematics continues on page 22. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Explain the rule for the sequence: 4, 7, 10, 13, 16... (Rule: Add 3, n-th term = 3n + 1).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 1.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 1.85]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 1.86]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 1.87]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 1.88]

Given the conditions of 1.2 Triangular Numbers and Geometric Arrangements, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

CHAPTER 2

LINES AND ANGLES

CBSE / NCERT Curriculum • Standard Grade 6 Mathematics

CORE TOPICS & LEARNING OBJECTIVES:

- 2.1 Points, Lines, Rays and Line Segments
- 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex
- 2.3 Measuring Angles with a Protractor
- 2.4 Intersecting Lines and Parallel Lines
- 2.5 Pairs of Angles: Complementary and Supplementary

1. Chapter Introduction & Conceptual Overview

In this chapter, students investigate the foundational principles of Lines and Angles. Mathematics is not merely about mechanical computations; it is about discovering relationships, finding symmetries, and applying structured reasoning to solve real-world problems.

NCERT Ganita Prakash introduces these concepts through tactile explorations, visual models, and guided inquiries. Students are encouraged to observe patterns carefully before formulating generalized rules.

KEY EXAMPLE 1.1 (WORKED DEMONSTRATION):

Problem Statement: Classify angles: 45° (Acute), 90° (Right), 135° (Obtuse), 180° (Straight), 240° (Reflex).

Step-by-step Solution: Follow standard NCERT methodology with full justifications.

NCERT DIDACTIC EXPLORATION CORNER:

Remember: Always verify your calculation by checking special cases and boundary values.

2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex

Mathematical analysis for Lines and Angles continues on page 24. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Identify parallel lines and transversal in a standard railway track diagram.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.2 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.5]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.6]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.7]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.8]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.3 Measuring Angles with a Protractor

Mathematical analysis for Lines and Angles continues on page 25. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Classify angles: 45° (Acute), 90° (Right), 135° (Obtuse), 180° (Straight), 240° (Reflex).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.3 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.9]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.10]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.11]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.12]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.4 Intersecting Lines and Parallel Lines

Mathematical analysis for Lines and Angles continues on page 26. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If two angles are complementary and one is 38°, find the other: 90° - 38° = 52°.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.13]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.14]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.15]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.16]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.5 Pairs of Angles: Complementary and Supplementary

Mathematical analysis for Lines and Angles continues on page 27. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Identify parallel lines and transversal in a standard railway track diagram.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.17]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.18]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.19]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.20]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.1 Points, Lines, Rays and Line Segments

Mathematical analysis for Lines and Angles continues on page 28. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Classify angles: 45° (Acute), 90° (Right), 135° (Obtuse), 180° (Straight), 240° (Reflex).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.21]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.22]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.23]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.24]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex

Mathematical analysis for Lines and Angles continues on page 29. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If two angles are complementary and one is 38°, find the other: 90° - 38° = 52°.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.25]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.26]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.27]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.28]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.3 Measuring Angles with a Protractor

Mathematical analysis for Lines and Angles continues on page 30. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Identify parallel lines and transversal in a standard railway track diagram.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.29]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.30]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.31]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.32]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.4 Intersecting Lines and Parallel Lines

Mathematical analysis for Lines and Angles continues on page 31. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Classify angles: 45° (Acute), 90° (Right), 135° (Obtuse), 180° (Straight), 240° (Reflex).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.33]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.34]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.35]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.36]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.5 Pairs of Angles: Complementary and Supplementary

Mathematical analysis for Lines and Angles continues on page 32. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If two angles are complementary and one is 38°, find the other: $90° - 38° = 52°$.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.37]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.38]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.39]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.40]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.1 Points, Lines, Rays and Line Segments

Mathematical analysis for Lines and Angles continues on page 33. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Identify parallel lines and transversal in a standard railway track diagram.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.41]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.42]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.43]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.44]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex

Mathematical analysis for Lines and Angles continues on page 34. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Classify angles: 45° (Acute), 90° (Right), 135° (Obtuse), 180° (Straight), 240° (Reflex).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.45]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.46]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.47]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.48]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.3 Measuring Angles with a Protractor

Mathematical analysis for Lines and Angles continues on page 35. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If two angles are complementary and one is 38°, find the other: 90° - 38° = 52°.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.49]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.50]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.51]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.52]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.4 Intersecting Lines and Parallel Lines

Mathematical analysis for Lines and Angles continues on page 36. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Identify parallel lines and transversal in a standard railway track diagram.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.53]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.54]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.55]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.56]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.5 Pairs of Angles: Complementary and Supplementary

Mathematical analysis for Lines and Angles continues on page 37. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Classify angles: 45° (Acute), 90° (Right), 135° (Obtuse), 180° (Straight), 240° (Reflex).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.57]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.58]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.59]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.60]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.1 Points, Lines, Rays and Line Segments

Mathematical analysis for Lines and Angles continues on page 38. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If two angles are complementary and one is 38°, find the other: 90° - 38° = 52°.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.61]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.62]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.63]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.64]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex

Mathematical analysis for Lines and Angles continues on page 39. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Identify parallel lines and transversal in a standard railway track diagram.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.65]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.66]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.67]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.68]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.3 Measuring Angles with a Protractor

Mathematical analysis for Lines and Angles continues on page 40. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Classify angles: 45° (Acute), 90° (Right), 135° (Obtuse), 180° (Straight), 240° (Reflex).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.69]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.70]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.71]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.72]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.4 Intersecting Lines and Parallel Lines

Mathematical analysis for Lines and Angles continues on page 41. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If two angles are complementary and one is 38°, find the other: 90° - 38° = 52°.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.73]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.74]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.75]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.76]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.5 Pairs of Angles: Complementary and Supplementary

Mathematical analysis for Lines and Angles continues on page 42. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Identify parallel lines and transversal in a standard railway track diagram.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.77]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.78]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.79]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.80]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.1 Points, Lines, Rays and Line Segments

Mathematical analysis for Lines and Angles continues on page 43. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Classify angles: 45° (Acute), 90° (Right), 135° (Obtuse), 180° (Straight), 240° (Reflex).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.81]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.82]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.83]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.84]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex

Mathematical analysis for Lines and Angles continues on page 44. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If two angles are complementary and one is 38°, find the other: $90° - 38° = 52°$.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.85]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.86]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.87]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.88]

Given the conditions of 2.2 Types of Angles: Acute, Right, Obtuse, Straight, Reflex, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.3 Measuring Angles with a Protractor

Mathematical analysis for Lines and Angles continues on page 45. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Identify parallel lines and transversal in a standard railway track diagram.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.89]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.90]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.91]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.92]

Given the conditions of 2.3 Measuring Angles with a Protractor, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.4 Intersecting Lines and Parallel Lines

Mathematical analysis for Lines and Angles continues on page 46. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Classify angles: 45° (Acute), 90° (Right), 135° (Obtuse), 180° (Straight), 240° (Reflex).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.93]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.94]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.95]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.96]

Given the conditions of 2.4 Intersecting Lines and Parallel Lines, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.5 Pairs of Angles: Complementary and Supplementary

Mathematical analysis for Lines and Angles continues on page 47. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If two angles are complementary and one is 38°, find the other: 90° - 38° = 52°.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.97]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.98]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.99]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.100]

Given the conditions of 2.5 Pairs of Angles: Complementary and Supplementary, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

2.1 Points, Lines, Rays and Line Segments

Mathematical analysis for Lines and Angles continues on page 48. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Identify parallel lines and transversal in a standard railway track diagram.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 2.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 2.101]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 2.102]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 2.103]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 2.104]

Given the conditions of 2.1 Points, Lines, Rays and Line Segments, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

CHAPTER 3

NUMBER PLAY

CBSE / NCERT Curriculum • Standard Grade 6 Mathematics

CORE TOPICS & LEARNING OBJECTIVES:

- 3.1 Magic Squares and Number Grids (3x3 and 4x4)
- 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11
- 3.3 Cryptarithms and Letter-Digit Puzzles
- 3.4 Factors and Multiples Investigation
- 3.5 Perfect Numbers and Palindromic Numbers

1. Chapter Introduction & Conceptual Overview

In this chapter, students investigate the foundational principles of Number Play. Mathematics is not merely about mechanical computations; it is about discovering relationships, finding symmetries, and applying structured reasoning to solve real-world problems.

NCERT Ganita Prakash introduces these concepts through tactile explorations, visual models, and guided inquiries. Students are encouraged to observe patterns carefully before formulating generalized rules.

KEY EXAMPLE 1.1 (WORKED DEMONSTRATION):

Problem Statement: Construct a 3x3 magic square with magic sum 15 using digits 1 to 9.

Step-by-step Solution: Follow standard NCERT methodology with full justifications.

NCERT DIDACTIC EXPLORATION CORNER:

Remember: Always verify your calculation by checking special cases and boundary values.

3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11

Mathematical analysis for Number Play continues on page 50. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find all factors of 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.2 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.5]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.6]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.7]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.8]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.3 Cryptarithms and Letter-Digit Puzzles

Mathematical analysis for Number Play continues on page 51. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 3x3 magic square with magic sum 15 using digits 1 to 9.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.3 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.9]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.10]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.11]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.12]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.4 Factors and Multiples Investigation

Mathematical analysis for Number Play continues on page 52. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Test whether 7,381,920 is divisible by 9 and 11 using divisibility tests.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.13]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.14]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.15]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.16]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.5 Perfect Numbers and Palindromic Numbers

Mathematical analysis for Number Play continues on page 53. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find all factors of 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.17]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.18]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.19]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.20]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.1 Magic Squares and Number Grids (3x3 and 4x4)

Mathematical analysis for Number Play continues on page 54. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 3x3 magic square with magic sum 15 using digits 1 to 9.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.21]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.22]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.23]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.24]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11

Mathematical analysis for Number Play continues on page 55. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Test whether 7,381,920 is divisible by 9 and 11 using divisibility tests.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.25]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.26]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.27]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.28]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.3 Cryptarithms and Letter-Digit Puzzles

Mathematical analysis for Number Play continues on page 56. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find all factors of 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.29]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.30]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.31]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.32]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.4 Factors and Multiples Investigation

Mathematical analysis for Number Play continues on page 57. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 3x3 magic square with magic sum 15 using digits 1 to 9.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.33]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.34]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.35]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.36]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.5 Perfect Numbers and Palindromic Numbers

Mathematical analysis for Number Play continues on page 58. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Test whether 7,381,920 is divisible by 9 and 11 using divisibility tests.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.37]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.38]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.39]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.40]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.1 Magic Squares and Number Grids (3x3 and 4x4)

Mathematical analysis for Number Play continues on page 59. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find all factors of 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.41]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.42]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.43]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.44]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11

Mathematical analysis for Number Play continues on page 60. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 3x3 magic square with magic sum 15 using digits 1 to 9.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.45]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.46]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.47]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.48]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.3 Cryptarithms and Letter-Digit Puzzles

Mathematical analysis for Number Play continues on page 61. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Test whether 7,381,920 is divisible by 9 and 11 using divisibility tests.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.49]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.50]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.51]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.52]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.4 Factors and Multiples Investigation

Mathematical analysis for Number Play continues on page 62. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find all factors of 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.53]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.54]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.55]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.56]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.5 Perfect Numbers and Palindromic Numbers

Mathematical analysis for Number Play continues on page 63. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 3x3 magic square with magic sum 15 using digits 1 to 9.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.57]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.58]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.59]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.60]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.1 Magic Squares and Number Grids (3x3 and 4x4)

Mathematical analysis for Number Play continues on page 64. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Test whether 7,381,920 is divisible by 9 and 11 using divisibility tests.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.61]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.62]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.63]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.64]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11

Mathematical analysis for Number Play continues on page 65. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find all factors of 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.65]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.66]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.67]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.68]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.3 Cryptarithms and Letter-Digit Puzzles

Mathematical analysis for Number Play continues on page 66. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 3x3 magic square with magic sum 15 using digits 1 to 9.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.69]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.70]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.71]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.72]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.4 Factors and Multiples Investigation

Mathematical analysis for Number Play continues on page 67. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Test whether 7,381,920 is divisible by 9 and 11 using divisibility tests.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.73]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.74]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.75]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.76]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.5 Perfect Numbers and Palindromic Numbers

Mathematical analysis for Number Play continues on page 68. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find all factors of 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.77]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.78]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.79]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.80]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.1 Magic Squares and Number Grids (3x3 and 4x4)

Mathematical analysis for Number Play continues on page 69. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 3x3 magic square with magic sum 15 using digits 1 to 9.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.81]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.82]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.83]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.84]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11

Mathematical analysis for Number Play continues on page 70. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Test whether 7,381,920 is divisible by 9 and 11 using divisibility tests.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.85]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.86]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.87]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.88]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.3 Cryptarithms and Letter-Digit Puzzles

Mathematical analysis for Number Play continues on page 71. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find all factors of 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.89]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.90]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.91]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.92]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.4 Factors and Multiples Investigation

Mathematical analysis for Number Play continues on page 72. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 3x3 magic square with magic sum 15 using digits 1 to 9.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.93]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.94]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.95]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.96]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.5 Perfect Numbers and Palindromic Numbers

Mathematical analysis for Number Play continues on page 73. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Test whether 7,381,920 is divisible by 9 and 11 using divisibility tests.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.97]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.98]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.99]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.100]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.1 Magic Squares and Number Grids (3x3 and 4x4)

Mathematical analysis for Number Play continues on page 74. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find all factors of 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.101]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.102]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.103]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.104]

Given the conditions of 3.1 Magic Squares and Number Grids (3x3 and 4x4), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11

Mathematical analysis for Number Play continues on page 75. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 3x3 magic square with magic sum 15 using digits 1 to 9.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.105]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.106]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.107]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.108]

Given the conditions of 3.2 Divisibility Rules for 2, 3, 4, 5, 6, 8, 9, 10, 11, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.3 Cryptarithms and Letter-Digit Puzzles

Mathematical analysis for Number Play continues on page 76. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Test whether 7,381,920 is divisible by 9 and 11 using divisibility tests.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.109]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.110]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.111]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.112]

Given the conditions of 3.3 Cryptarithms and Letter-Digit Puzzles, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.4 Factors and Multiples Investigation

Mathematical analysis for Number Play continues on page 77. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find all factors of 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.113]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.114]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.115]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.116]

Given the conditions of 3.4 Factors and Multiples Investigation, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

3.5 Perfect Numbers and Palindromic Numbers

Mathematical analysis for Number Play continues on page 78. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 3x3 magic square with magic sum 15 using digits 1 to 9.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 3.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 3.117]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 3.118]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 3.119]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 3.120]

Given the conditions of 3.5 Perfect Numbers and Palindromic Numbers, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

CHAPTER 4

DATA HANDLING AND PRESENTATION

CBSE / NCERT Curriculum • Standard Grade 6 Mathematics

CORE TOPICS & LEARNING OBJECTIVES:

- 4.1 Collection and Organization of Raw Data
- 4.2 Tally Marks and Frequency Tables
- 4.3 Pictographs with Scale Keys
- 4.4 Bar Graphs: Vertical and Horizontal Representations
- 4.5 Interpreting Trends and Drawing Inferences

1. Chapter Introduction & Conceptual Overview

In this chapter, students investigate the foundational principles of Data Handling and Presentation. Mathematics is not merely about mechanical computations; it is about discovering relationships, finding symmetries, and applying structured reasoning to solve real-world problems.

NCERT Ganita Prakash introduces these concepts through tactile explorations, visual models, and guided inquiries. Students are encouraged to observe patterns carefully before formulating generalized rules.

KEY EXAMPLE 1.1 (WORKED DEMONSTRATION):

Problem Statement: Organize the test scores of 30 students into a frequency table with intervals.

Step-by-step Solution: Follow standard NCERT methodology with full justifications.

NCERT DIDACTIC EXPLORATION CORNER:

Remember: Always verify your calculation by checking special cases and boundary values.

4.2 Tally Marks and Frequency Tables

Mathematical analysis for Data Handling and Presentation continues on page 80. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a bar graph showing favorite sports of Class 6 students.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.2 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.5]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.6]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.7]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.8]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.3 Pictographs with Scale Keys

Mathematical analysis for Data Handling and Presentation continues on page 81. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Organize the test scores of 30 students into a frequency table with intervals.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.3 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.9]

Given the conditions of 4.3 Pictographs with Scale Keys, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.10]

Given the conditions of 4.3 Pictographs with Scale Keys, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.11]

Given the conditions of 4.3 Pictographs with Scale Keys, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.12]

Given the conditions of 4.3 Pictographs with Scale Keys, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.4 Bar Graphs: Vertical and Horizontal Representations

Mathematical analysis for Data Handling and Presentation continues on page 82. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw a pictograph where 1 symbol represents 5 books read by library club.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.13]

Given the conditions of 4.4 Bar Graphs: Vertical and Horizontal Representations, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.14]

Given the conditions of 4.4 Bar Graphs: Vertical and Horizontal Representations, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.15]

Given the conditions of 4.4 Bar Graphs: Vertical and Horizontal Representations, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.16]

Given the conditions of 4.4 Bar Graphs: Vertical and Horizontal Representations, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.5 Interpreting Trends and Drawing Inferences

Mathematical analysis for Data Handling and Presentation continues on page 83. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a bar graph showing favorite sports of Class 6 students.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.17]

Given the conditions of 4.5 Interpreting Trends and Drawing Inferences, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.18]

Given the conditions of 4.5 Interpreting Trends and Drawing Inferences, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.19]

Given the conditions of 4.5 Interpreting Trends and Drawing Inferences, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.20]

Given the conditions of 4.5 Interpreting Trends and Drawing Inferences, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.1 Collection and Organization of Raw Data

Mathematical analysis for Data Handling and Presentation continues on page 84. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Organize the test scores of 30 students into a frequency table with intervals.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.21]

Given the conditions of 4.1 Collection and Organization of Raw Data, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.22]

Given the conditions of 4.1 Collection and Organization of Raw Data, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.23]

Given the conditions of 4.1 Collection and Organization of Raw Data, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.24]

Given the conditions of 4.1 Collection and Organization of Raw Data, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.2 Tally Marks and Frequency Tables

Mathematical analysis for Data Handling and Presentation continues on page 85. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw a pictograph where 1 symbol represents 5 books read by library club.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.25]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.26]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.27]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.28]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.3 Pictographs with Scale Keys

Mathematical analysis for Data Handling and Presentation continues on page 86. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a bar graph showing favorite sports of Class 6 students.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.29]

Given the conditions of 4.3 Pictographs with Scale Keys, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.30]

Given the conditions of 4.3 Pictographs with Scale Keys, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.31]

Given the conditions of 4.3 Pictographs with Scale Keys, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.32]

Given the conditions of 4.3 Pictographs with Scale Keys, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.4 Bar Graphs: Vertical and Horizontal Representations

Mathematical analysis for Data Handling and Presentation continues on page 87. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Organize the test scores of 30 students into a frequency table with intervals.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.33]

Given the conditions of 4.4 Bar Graphs: Vertical and Horizontal Representations, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.34]

Given the conditions of 4.4 Bar Graphs: Vertical and Horizontal Representations, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.35]

Given the conditions of 4.4 Bar Graphs: Vertical and Horizontal Representations, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.36]

Given the conditions of 4.4 Bar Graphs: Vertical and Horizontal Representations, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.5 Interpreting Trends and Drawing Inferences

Mathematical analysis for Data Handling and Presentation continues on page 88. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw a pictograph where 1 symbol represents 5 books read by library club.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.37]

Given the conditions of 4.5 Interpreting Trends and Drawing Inferences, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.38]

Given the conditions of 4.5 Interpreting Trends and Drawing Inferences, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.39]

Given the conditions of 4.5 Interpreting Trends and Drawing Inferences, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.40]

Given the conditions of 4.5 Interpreting Trends and Drawing Inferences, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.1 Collection and Organization of Raw Data

Mathematical analysis for Data Handling and Presentation continues on page 89. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a bar graph showing favorite sports of Class 6 students.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.41]

Given the conditions of 4.1 Collection and Organization of Raw Data, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.42]

Given the conditions of 4.1 Collection and Organization of Raw Data, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.43]

Given the conditions of 4.1 Collection and Organization of Raw Data, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.44]

Given the conditions of 4.1 Collection and Organization of Raw Data, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.2 Tally Marks and Frequency Tables

Mathematical analysis for Data Handling and Presentation continues on page 90. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Organize the test scores of 30 students into a frequency table with intervals.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.45]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.46]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.47]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.48]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.3 Pictographs with Scale Keys

Mathematical analysis for Data Handling and Presentation continues on page 91. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw a pictograph where 1 symbol represents 5 books read by library club.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.49]

Given the conditions of 4.3 Pictographs with Scale Keys, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.50]

Given the conditions of 4.3 Pictographs with Scale Keys, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.51]

Given the conditions of 4.3 Pictographs with Scale Keys, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.52]

Given the conditions of 4.3 Pictographs with Scale Keys, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.4 Bar Graphs: Vertical and Horizontal Representations

Mathematical analysis for Data Handling and Presentation continues on page 92. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a bar graph showing favorite sports of Class 6 students.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.53]

Given the conditions of 4.4 Bar Graphs: Vertical and Horizontal Representations, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.54]

Given the conditions of 4.4 Bar Graphs: Vertical and Horizontal Representations, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.55]

Given the conditions of 4.4 Bar Graphs: Vertical and Horizontal Representations, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.56]

Given the conditions of 4.4 Bar Graphs: Vertical and Horizontal Representations, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.5 Interpreting Trends and Drawing Inferences

Mathematical analysis for Data Handling and Presentation continues on page 93. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Organize the test scores of 30 students into a frequency table with intervals.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.57]

Given the conditions of 4.5 Interpreting Trends and Drawing Inferences, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.58]

Given the conditions of 4.5 Interpreting Trends and Drawing Inferences, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.59]

Given the conditions of 4.5 Interpreting Trends and Drawing Inferences, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.60]

Given the conditions of 4.5 Interpreting Trends and Drawing Inferences, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.1 Collection and Organization of Raw Data

Mathematical analysis for Data Handling and Presentation continues on page 94. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw a pictograph where 1 symbol represents 5 books read by library club.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.61]

Given the conditions of 4.1 Collection and Organization of Raw Data, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.62]

Given the conditions of 4.1 Collection and Organization of Raw Data, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.63]

Given the conditions of 4.1 Collection and Organization of Raw Data, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.64]

Given the conditions of 4.1 Collection and Organization of Raw Data, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.2 Tally Marks and Frequency Tables

Mathematical analysis for Data Handling and Presentation continues on page 95. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a bar graph showing favorite sports of Class 6 students.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.65]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.66]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.67]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.68]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.3 Pictographs with Scale Keys

Mathematical analysis for Data Handling and Presentation continues on page 96. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Organize the test scores of 30 students into a frequency table with intervals.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.69]

Given the conditions of 4.3 Pictographs with Scale Keys, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.70]

Given the conditions of 4.3 Pictographs with Scale Keys, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.71]

Given the conditions of 4.3 Pictographs with Scale Keys, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.72]

Given the conditions of 4.3 Pictographs with Scale Keys, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.4 Bar Graphs: Vertical and Horizontal Representations

Mathematical analysis for Data Handling and Presentation continues on page 97. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw a pictograph where 1 symbol represents 5 books read by library club.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.73]

Given the conditions of 4.4 Bar Graphs: Vertical and Horizontal Representations, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.74]

Given the conditions of 4.4 Bar Graphs: Vertical and Horizontal Representations, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.75]

Given the conditions of 4.4 Bar Graphs: Vertical and Horizontal Representations, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.76]

Given the conditions of 4.4 Bar Graphs: Vertical and Horizontal Representations, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.5 Interpreting Trends and Drawing Inferences

Mathematical analysis for Data Handling and Presentation continues on page 98. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a bar graph showing favorite sports of Class 6 students.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.77]

Given the conditions of 4.5 Interpreting Trends and Drawing Inferences, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.78]

Given the conditions of 4.5 Interpreting Trends and Drawing Inferences, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.79]

Given the conditions of 4.5 Interpreting Trends and Drawing Inferences, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.80]

Given the conditions of 4.5 Interpreting Trends and Drawing Inferences, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.1 Collection and Organization of Raw Data

Mathematical analysis for Data Handling and Presentation continues on page 99. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Organize the test scores of 30 students into a frequency table with intervals.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.81]

Given the conditions of 4.1 Collection and Organization of Raw Data, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.82]

Given the conditions of 4.1 Collection and Organization of Raw Data, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.83]

Given the conditions of 4.1 Collection and Organization of Raw Data, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.84]

Given the conditions of 4.1 Collection and Organization of Raw Data, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

4.2 Tally Marks and Frequency Tables

Mathematical analysis for Data Handling and Presentation continues on page 100. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw a pictograph where 1 symbol represents 5 books read by library club.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 4.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 4.85]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 4.86]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 4.87]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 4.88]

Given the conditions of 4.2 Tally Marks and Frequency Tables, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

CHAPTER 5

PRIME TIME

CBSE / NCERT Curriculum • Standard Grade 6 Mathematics

CORE TOPICS & LEARNING OBJECTIVES:

- 5.1 Prime Numbers and Composite Numbers
- 5.2 Sieve of Eratosthenes (1 to 100)
- 5.3 Prime Factorization using Factor Trees and Continuous Division
- 5.4 Highest Common Factor (HCF / GCD)
- 5.5 Lowest Common Multiple (LCM) and Word Problems

1. Chapter Introduction & Conceptual Overview

In this chapter, students investigate the foundational principles of Prime Time. Mathematics is not merely about mechanical computations; it is about discovering relationships, finding symmetries, and applying structured reasoning to solve real-world problems.

NCERT Ganita Prakash introduces these concepts through tactile explorations, visual models, and guided inquiries. Students are encouraged to observe patterns carefully before formulating generalized rules.

KEY EXAMPLE 1.1 (WORKED DEMONSTRATION):

Problem Statement: List all prime numbers between 50 and 80: 53, 59, 61, 67, 71, 73, 79.

Step-by-step Solution: Follow standard NCERT methodology with full justifications.

NCERT DIDACTIC EXPLORATION CORNER:

Remember: Always verify your calculation by checking special cases and boundary values.

5.2 Sieve of Eratosthenes (1 to 100)

Mathematical analysis for Prime Time continues on page 102. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Three bells toll at intervals of 9, 12, 15 minutes. Find when they toll together (LCM = 180 min = 3 hrs).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.2 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.5]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.6]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.7]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.8]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.3 Prime Factorization using Factor Trees and Continuous Division

Mathematical analysis for Prime Time continues on page 103. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: List all prime numbers between 50 and 80: 53, 59, 61, 67, 71, 73, 79.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.3 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.9]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.10]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.11]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.12]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.4 Highest Common Factor (HCF / GCD)

Mathematical analysis for Prime Time continues on page 104. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the HCF of 84 and 120 using prime factorization: $84 = 2^2 \times 3 \times 7$, $120 = 2^3 \times 3 \times 5$. HCF = 12.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.13]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.14]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.15]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.16]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.5 Lowest Common Multiple (LCM) and Word Problems

Mathematical analysis for Prime Time continues on page 105. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Three bells toll at intervals of 9, 12, 15 minutes. Find when they toll together (LCM = 180 min = 3 hrs).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.17]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.18]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.19]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.20]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.1 Prime Numbers and Composite Numbers

Mathematical analysis for Prime Time continues on page 106. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: List all prime numbers between 50 and 80: 53, 59, 61, 67, 71, 73, 79.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.21]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.22]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.23]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.24]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.2 Sieve of Eratosthenes (1 to 100)

Mathematical analysis for Prime Time continues on page 107. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the HCF of 84 and 120 using prime factorization: 84 = 2^2 x 3 x 7, 120 = 2^3 x 3 x 5. HCF = 12.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.25]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.26]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.27]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.28]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.3 Prime Factorization using Factor Trees and Continuous Division

Mathematical analysis for Prime Time continues on page 108. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Three bells toll at intervals of 9, 12, 15 minutes. Find when they toll together (LCM = 180 min = 3 hrs).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.29]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.30]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.31]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.32]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.4 Highest Common Factor (HCF / GCD)

Mathematical analysis for Prime Time continues on page 109. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: List all prime numbers between 50 and 80: 53, 59, 61, 67, 71, 73, 79.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.33]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.34]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.35]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.36]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.5 Lowest Common Multiple (LCM) and Word Problems

Mathematical analysis for Prime Time continues on page 110. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the HCF of 84 and 120 using prime factorization: $84 = 2^2 \times 3 \times 7$, $120 = 2^3 \times 3 \times 5$. HCF = 12.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.37]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.38]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.39]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.40]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.1 Prime Numbers and Composite Numbers

Mathematical analysis for Prime Time continues on page 111. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Three bells toll at intervals of 9, 12, 15 minutes. Find when they toll together (LCM = 180 min = 3 hrs).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.41]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.42]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.43]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.44]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.2 Sieve of Eratosthenes (1 to 100)

Mathematical analysis for Prime Time continues on page 112. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: List all prime numbers between 50 and 80: 53, 59, 61, 67, 71, 73, 79.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.45]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.46]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.47]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.48]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.3 Prime Factorization using Factor Trees and Continuous Division

Mathematical analysis for Prime Time continues on page 113. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the HCF of 84 and 120 using prime factorization: 84 = 2^2 x 3 x 7, 120 = 2^3 x 3 x 5. HCF = 12.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.49]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.50]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.51]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.52]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.4 Highest Common Factor (HCF / GCD)

Mathematical analysis for Prime Time continues on page 114. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Three bells toll at intervals of 9, 12, 15 minutes. Find when they toll together (LCM = 180 min = 3 hrs).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.53]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.54]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.55]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.56]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.5 Lowest Common Multiple (LCM) and Word Problems

Mathematical analysis for Prime Time continues on page 115. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: List all prime numbers between 50 and 80: 53, 59, 61, 67, 71, 73, 79.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.57]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.58]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.59]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.60]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.1 Prime Numbers and Composite Numbers

Mathematical analysis for Prime Time continues on page 116. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the HCF of 84 and 120 using prime factorization: $84 = 2^2 \times 3 \times 7$, $120 = 2^3 \times 3 \times 5$. HCF = 12.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.61]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.62]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.63]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.64]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.2 Sieve of Eratosthenes (1 to 100)

Mathematical analysis for Prime Time continues on page 117. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Three bells toll at intervals of 9, 12, 15 minutes. Find when they toll together (LCM = 180 min = 3 hrs).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.65]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.66]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.67]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.68]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.3 Prime Factorization using Factor Trees and Continuous Division

Mathematical analysis for Prime Time continues on page 118. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: List all prime numbers between 50 and 80: 53, 59, 61, 67, 71, 73, 79.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.69]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.70]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.71]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.72]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.4 Highest Common Factor (HCF / GCD)

Mathematical analysis for Prime Time continues on page 119. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the HCF of 84 and 120 using prime factorization: $84 = 2^2 \times 3 \times 7$, $120 = 2^3 \times 3 \times 5$. HCF = 12.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.73]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.74]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.75]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.76]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.5 Lowest Common Multiple (LCM) and Word Problems

Mathematical analysis for Prime Time continues on page 120. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Three bells toll at intervals of 9, 12, 15 minutes. Find when they toll together (LCM = 180 min = 3 hrs).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.77]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.78]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.79]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.80]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.1 Prime Numbers and Composite Numbers

Mathematical analysis for Prime Time continues on page 121. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: List all prime numbers between 50 and 80: 53, 59, 61, 67, 71, 73, 79.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.81]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.82]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.83]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.84]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.2 Sieve of Eratosthenes (1 to 100)

Mathematical analysis for Prime Time continues on page 122. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the HCF of 84 and 120 using prime factorization: 84 = 2^2 x 3 x 7, 120 = 2^3 x 3 x 5. HCF = 12.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.85]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.86]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.87]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.88]

Given the conditions of 5.2 Sieve of Eratosthenes (1 to 100), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.3 Prime Factorization using Factor Trees and Continuous Division

Mathematical analysis for Prime Time continues on page 123. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Three bells toll at intervals of 9, 12, 15 minutes. Find when they toll together (LCM = 180 min = 3 hrs).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.89]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.90]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.91]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.92]

Given the conditions of 5.3 Prime Factorization using Factor Trees and Continuous Division, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.4 Highest Common Factor (HCF / GCD)

Mathematical analysis for Prime Time continues on page 124. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: List all prime numbers between 50 and 80: 53, 59, 61, 67, 71, 73, 79.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.93]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.94]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.95]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.96]

Given the conditions of 5.4 Highest Common Factor (HCF / GCD), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.5 Lowest Common Multiple (LCM) and Word Problems

Mathematical analysis for Prime Time continues on page 125. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the HCF of 84 and 120 using prime factorization: 84 = 2^2 x 3 x 7, 120 = 2^3 x 3 x 5. HCF = 12.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.97]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.98]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.99]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.100]

Given the conditions of 5.5 Lowest Common Multiple (LCM) and Word Problems, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

5.1 Prime Numbers and Composite Numbers

Mathematical analysis for Prime Time continues on page 126. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Three bells toll at intervals of 9, 12, 15 minutes. Find when they toll together (LCM = 180 min = 3 hrs).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 5.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 5.101]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 5.102]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 5.103]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 5.104]

Given the conditions of 5.1 Prime Numbers and Composite Numbers, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

CHAPTER 6

PERIMETER AND AREA

CBSE / NCERT Curriculum • Standard Grade 6 Mathematics

CORE TOPICS & LEARNING OBJECTIVES:

- 6.1 Concept of Perimeter as Boundary Length
- 6.2 Perimeter of Rectangles, Squares and Regular Polygons
- 6.3 Concept of Area using Unit Square Grids
- 6.4 Area Formulae: Rectangle (l x b) and Square (side^2)
- 6.5 Real-life Problems on Fencing, Flooring and Tiling

1. Chapter Introduction & Conceptual Overview

In this chapter, students investigate the foundational principles of Perimeter and Area. Mathematics is not merely about mechanical computations; it is about discovering relationships, finding symmetries, and applying structured reasoning to solve real-world problems.

NCERT Ganita Prakash introduces these concepts through tactile explorations, visual models, and guided inquiries. Students are encouraged to observe patterns carefully before formulating generalized rules.

KEY EXAMPLE 1.1 (WORKED DEMONSTRATION):

Problem Statement: Find the perimeter of a rectangular park of length 150 m and breadth 80 m: 2(150 + 80) = 460 m.

Step-by-step Solution: Follow standard NCERT methodology with full justifications.

NCERT DIDACTIC EXPLORATION CORNER:

Remember: Always verify your calculation by checking special cases and boundary values.

6.2 Perimeter of Rectangles, Squares and Regular Polygons

Mathematical analysis for Perimeter and Area continues on page 128. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: How many 20 cm x 10 cm tiles are needed to cover a 4 m x 3 m floor? Total = 600 tiles.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.2 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.5]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.6]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.7]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.8]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.3 Concept of Area using Unit Square Grids

Mathematical analysis for Perimeter and Area continues on page 129. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the perimeter of a rectangular park of length 150 m and breadth 80 m: $2(150 + 80) = 460$ m.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.3 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.9]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.10]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.11]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.12]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.4 Area Formulae: Rectangle (l x b) and Square (side^2)

Mathematical analysis for Perimeter and Area continues on page 130. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the area of a square tile with side 25 cm: 25 x 25 = 625 cm^2.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.13]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.14]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.15]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.16]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.5 Real-life Problems on Fencing, Flooring and Tiling

Mathematical analysis for Perimeter and Area continues on page 131. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: How many 20 cm x 10 cm tiles are needed to cover a 4 m x 3 m floor? Total = 600 tiles.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.17]

Given the conditions of 6.5 Real-life Problems on Fencing, Flooring and Tiling, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.18]

Given the conditions of 6.5 Real-life Problems on Fencing, Flooring and Tiling, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.19]

Given the conditions of 6.5 Real-life Problems on Fencing, Flooring and Tiling, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.20]

Given the conditions of 6.5 Real-life Problems on Fencing, Flooring and Tiling, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.1 Concept of Perimeter as Boundary Length

Mathematical analysis for Perimeter and Area continues on page 132. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the perimeter of a rectangular park of length 150 m and breadth 80 m: $2(150 + 80) = 460$ m.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.21]

Given the conditions of 6.1 Concept of Perimeter as Boundary Length, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.22]

Given the conditions of 6.1 Concept of Perimeter as Boundary Length, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.23]

Given the conditions of 6.1 Concept of Perimeter as Boundary Length, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.24]

Given the conditions of 6.1 Concept of Perimeter as Boundary Length, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.2 Perimeter of Rectangles, Squares and Regular Polygons

Mathematical analysis for Perimeter and Area continues on page 133. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the area of a square tile with side 25 cm: 25 x 25 = 625 cm^2.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.25]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.26]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.27]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.28]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.3 Concept of Area using Unit Square Grids

Mathematical analysis for Perimeter and Area continues on page 134. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: How many 20 cm x 10 cm tiles are needed to cover a 4 m x 3 m floor? Total = 600 tiles.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.29]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.30]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.31]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.32]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.4 Area Formulae: Rectangle (l x b) and Square (side^2)

Mathematical analysis for Perimeter and Area continues on page 135. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the perimeter of a rectangular park of length 150 m and breadth 80 m: 2(150 + 80) = 460 m.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.33]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.34]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.35]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.36]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.5 Real-life Problems on Fencing, Flooring and Tiling

Mathematical analysis for Perimeter and Area continues on page 136. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the area of a square tile with side 25 cm: 25 x 25 = 625 cm^2.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.37]

Given the conditions of 6.5 Real-life Problems on Fencing, Flooring and Tiling, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.38]

Given the conditions of 6.5 Real-life Problems on Fencing, Flooring and Tiling, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.39]

Given the conditions of 6.5 Real-life Problems on Fencing, Flooring and Tiling, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.40]

Given the conditions of 6.5 Real-life Problems on Fencing, Flooring and Tiling, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.1 Concept of Perimeter as Boundary Length

Mathematical analysis for Perimeter and Area continues on page 137. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: How many 20 cm x 10 cm tiles are needed to cover a 4 m x 3 m floor? Total = 600 tiles.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.41]

Given the conditions of 6.1 Concept of Perimeter as Boundary Length, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.42]

Given the conditions of 6.1 Concept of Perimeter as Boundary Length, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.43]

Given the conditions of 6.1 Concept of Perimeter as Boundary Length, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.44]

Given the conditions of 6.1 Concept of Perimeter as Boundary Length, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.2 Perimeter of Rectangles, Squares and Regular Polygons

Mathematical analysis for Perimeter and Area continues on page 138. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the perimeter of a rectangular park of length 150 m and breadth 80 m: $2(150 + 80) = 460$ m.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.45]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.46]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.47]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.48]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.3 Concept of Area using Unit Square Grids

Mathematical analysis for Perimeter and Area continues on page 139. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the area of a square tile with side 25 cm: 25 x 25 = 625 cm^2.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.49]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.50]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.51]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.52]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.4 Area Formulae: Rectangle (l x b) and Square (side^2)

Mathematical analysis for Perimeter and Area continues on page 140. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: How many 20 cm x 10 cm tiles are needed to cover a 4 m x 3 m floor? Total = 600 tiles.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.53]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.54]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.55]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.56]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.5 Real-life Problems on Fencing, Flooring and Tiling

Mathematical analysis for Perimeter and Area continues on page 141. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the perimeter of a rectangular park of length 150 m and breadth 80 m: 2(150 + 80) = 460 m.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.57]

Given the conditions of 6.5 Real-life Problems on Fencing, Flooring and Tiling, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.58]

Given the conditions of 6.5 Real-life Problems on Fencing, Flooring and Tiling, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.59]

Given the conditions of 6.5 Real-life Problems on Fencing, Flooring and Tiling, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.60]

Given the conditions of 6.5 Real-life Problems on Fencing, Flooring and Tiling, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.1 Concept of Perimeter as Boundary Length

Mathematical analysis for Perimeter and Area continues on page 142. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the area of a square tile with side 25 cm: 25 x 25 = 625 cm^2.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.61]

Given the conditions of 6.1 Concept of Perimeter as Boundary Length, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.62]

Given the conditions of 6.1 Concept of Perimeter as Boundary Length, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.63]

Given the conditions of 6.1 Concept of Perimeter as Boundary Length, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.64]

Given the conditions of 6.1 Concept of Perimeter as Boundary Length, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.2 Perimeter of Rectangles, Squares and Regular Polygons

Mathematical analysis for Perimeter and Area continues on page 143. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: How many 20 cm x 10 cm tiles are needed to cover a 4 m x 3 m floor? Total = 600 tiles.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.65]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.66]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.67]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.68]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.3 Concept of Area using Unit Square Grids

Mathematical analysis for Perimeter and Area continues on page 144. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the perimeter of a rectangular park of length 150 m and breadth 80 m: 2(150 + 80) = 460 m.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.69]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.70]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.71]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.72]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.4 Area Formulae: Rectangle (l x b) and Square (side^2)

Mathematical analysis for Perimeter and Area continues on page 145. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the area of a square tile with side 25 cm: 25 x 25 = 625 cm^2.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.73]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.74]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.75]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.76]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.5 Real-life Problems on Fencing, Flooring and Tiling

Mathematical analysis for Perimeter and Area continues on page 146. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: How many 20 cm x 10 cm tiles are needed to cover a 4 m x 3 m floor? Total = 600 tiles.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.77]

Given the conditions of 6.5 Real-life Problems on Fencing, Flooring and Tiling, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.78]

Given the conditions of 6.5 Real-life Problems on Fencing, Flooring and Tiling, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.79]

Given the conditions of 6.5 Real-life Problems on Fencing, Flooring and Tiling, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.80]

Given the conditions of 6.5 Real-life Problems on Fencing, Flooring and Tiling, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.1 Concept of Perimeter as Boundary Length

Mathematical analysis for Perimeter and Area continues on page 147. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the perimeter of a rectangular park of length 150 m and breadth 80 m: $2(150 + 80) = 460$ m.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.81]

Given the conditions of 6.1 Concept of Perimeter as Boundary Length, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.82]

Given the conditions of 6.1 Concept of Perimeter as Boundary Length, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.83]

Given the conditions of 6.1 Concept of Perimeter as Boundary Length, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.84]

Given the conditions of 6.1 Concept of Perimeter as Boundary Length, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.2 Perimeter of Rectangles, Squares and Regular Polygons

Mathematical analysis for Perimeter and Area continues on page 148. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the area of a square tile with side 25 cm: 25 x 25 = 625 cm^2.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.85]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.86]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.87]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.88]

Given the conditions of 6.2 Perimeter of Rectangles, Squares and Regular Polygons, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.3 Concept of Area using Unit Square Grids

Mathematical analysis for Perimeter and Area continues on page 149. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: How many 20 cm x 10 cm tiles are needed to cover a 4 m x 3 m floor? Total = 600 tiles.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.89]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.90]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.91]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.92]

Given the conditions of 6.3 Concept of Area using Unit Square Grids, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

6.4 Area Formulae: Rectangle (l x b) and Square (side^2)

Mathematical analysis for Perimeter and Area continues on page 150. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the perimeter of a rectangular park of length 150 m and breadth 80 m: 2(150 + 80) = 460 m.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 6.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 6.93]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 6.94]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 6.95]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 6.96]

Given the conditions of 6.4 Area Formulae: Rectangle (l x b) and Square (side^2), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

CHAPTER 7

FRACTIONS

CBSE / NCERT Curriculum • Standard Grade 6 Mathematics

CORE TOPICS & LEARNING OBJECTIVES:

- 7.1 Understanding Fractions as Part of a Whole and Collection
- 7.2 Fractions on the Number Line
- 7.3 Proper, Improper and Mixed Fractions
- 7.4 Equivalent Fractions and Simplest / Lowest Form
- 7.5 Comparing, Adding and Subtracting Fractions

1. Chapter Introduction & Conceptual Overview

In this chapter, students investigate the foundational principles of Fractions. Mathematics is not merely about mechanical computations; it is about discovering relationships, finding symmetries, and applying structured reasoning to solve real-world problems.

NCERT Ganita Prakash introduces these concepts through tactile explorations, visual models, and guided inquiries. Students are encouraged to observe patterns carefully before formulating generalized rules.

KEY EXAMPLE 1.1 (WORKED DEMONSTRATION):

Problem Statement: Convert 17/5 into a mixed fraction: 3 2/5.

Step-by-step Solution: Follow standard NCERT methodology with full justifications.

NCERT DIDACTIC EXPLORATION CORNER:

Remember: Always verify your calculation by checking special cases and boundary values.

7.2 Fractions on the Number Line

Mathematical analysis for Fractions continues on page 152. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Compute 3/8 + 5/12: LCM(8,12) = 24. (9 + 10)/24 = 19/24.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.2 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.5]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.6]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.7]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.8]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.3 Proper, Improper and Mixed Fractions

Mathematical analysis for Fractions continues on page 153. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Convert 17/5 into a mixed fraction: 3 2/5.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.3 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.9]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.10]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.11]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.12]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.4 Equivalent Fractions and Simplest / Lowest Form

Mathematical analysis for Fractions continues on page 154. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find 3 equivalent fractions for 4/7: 8/14, 12/21, 16/28.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.13]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.14]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.15]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.16]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.5 Comparing, Adding and Subtracting Fractions

Mathematical analysis for Fractions continues on page 155. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Compute $3/8 + 5/12$: $\mathrm{LCM}(8,12) = 24$. $(9 + 10)/24 = 19/24$.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.17]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.18]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.19]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.20]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.1 Understanding Fractions as Part of a Whole and Collection

Mathematical analysis for Fractions continues on page 156. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Convert 17/5 into a mixed fraction: 3 2/5.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.21]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.22]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.23]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.24]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.2 Fractions on the Number Line

Mathematical analysis for Fractions continues on page 157. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find 3 equivalent fractions for 4/7: 8/14, 12/21, 16/28.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.25]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.26]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.27]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.28]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.3 Proper, Improper and Mixed Fractions

Mathematical analysis for Fractions continues on page 158. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Compute $3/8 + 5/12$: $LCM(8,12) = 24$. $(9 + 10)/24 = 19/24$.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.29]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.30]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.31]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.32]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.4 Equivalent Fractions and Simplest / Lowest Form

Mathematical analysis for Fractions continues on page 159. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Convert 17/5 into a mixed fraction: 3 2/5.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.33]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.34]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.35]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.36]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.5 Comparing, Adding and Subtracting Fractions

Mathematical analysis for Fractions continues on page 160. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find 3 equivalent fractions for 4/7: 8/14, 12/21, 16/28.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.37]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.38]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.39]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.40]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.1 Understanding Fractions as Part of a Whole and Collection

Mathematical analysis for Fractions continues on page 161. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Compute $3/8 + 5/12$: $\text{LCM}(8,12) = 24$. $(9 + 10)/24 = 19/24$.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.41]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.42]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.43]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.44]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.2 Fractions on the Number Line

Mathematical analysis for Fractions continues on page 162. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Convert 17/5 into a mixed fraction: 3 2/5.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.45]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.46]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.47]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.48]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.3 Proper, Improper and Mixed Fractions

Mathematical analysis for Fractions continues on page 163. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find 3 equivalent fractions for 4/7: 8/14, 12/21, 16/28.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.49]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.50]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.51]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.52]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.4 Equivalent Fractions and Simplest / Lowest Form

Mathematical analysis for Fractions continues on page 164. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Compute 3/8 + 5/12: LCM(8,12) = 24. (9 + 10)/24 = 19/24.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.53]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.54]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.55]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.56]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.5 Comparing, Adding and Subtracting Fractions

Mathematical analysis for Fractions continues on page 165. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Convert 17/5 into a mixed fraction: 3 2/5.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.57]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.58]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.59]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.60]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.1 Understanding Fractions as Part of a Whole and Collection

Mathematical analysis for Fractions continues on page 166. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find 3 equivalent fractions for 4/7: 8/14, 12/21, 16/28.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.61]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.62]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.63]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.64]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.2 Fractions on the Number Line

Mathematical analysis for Fractions continues on page 167. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Compute 3/8 + 5/12: LCM(8,12) = 24. (9 + 10)/24 = 19/24.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.65]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.66]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.67]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.68]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.3 Proper, Improper and Mixed Fractions

Mathematical analysis for Fractions continues on page 168. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Convert 17/5 into a mixed fraction: 3 2/5.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.69]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.70]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.71]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.72]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.4 Equivalent Fractions and Simplest / Lowest Form

Mathematical analysis for Fractions continues on page 169. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find 3 equivalent fractions for 4/7: 8/14, 12/21, 16/28.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.73]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.74]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.75]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.76]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.5 Comparing, Adding and Subtracting Fractions

Mathematical analysis for Fractions continues on page 170. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Compute 3/8 + 5/12: LCM(8,12) = 24. (9 + 10)/24 = 19/24.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.77]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.78]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.79]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.80]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.1 Understanding Fractions as Part of a Whole and Collection

Mathematical analysis for Fractions continues on page 171. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Convert 17/5 into a mixed fraction: 3 2/5.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.81]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.82]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.83]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.84]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.2 Fractions on the Number Line

Mathematical analysis for Fractions continues on page 172. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find 3 equivalent fractions for 4/7: 8/14, 12/21, 16/28.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.85]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.86]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.87]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.88]

Given the conditions of 7.2 Fractions on the Number Line, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.3 Proper, Improper and Mixed Fractions

Mathematical analysis for Fractions continues on page 173. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Compute $3/8 + 5/12$: $LCM(8,12) = 24$. $(9 + 10)/24 = 19/24$.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.89]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.90]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.91]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.92]

Given the conditions of 7.3 Proper, Improper and Mixed Fractions, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.4 Equivalent Fractions and Simplest / Lowest Form

Mathematical analysis for Fractions continues on page 174. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Convert 17/5 into a mixed fraction: 3 2/5.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.93]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.94]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.95]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.96]

Given the conditions of 7.4 Equivalent Fractions and Simplest / Lowest Form, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.5 Comparing, Adding and Subtracting Fractions

Mathematical analysis for Fractions continues on page 175. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find 3 equivalent fractions for 4/7: 8/14, 12/21, 16/28.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.97]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.98]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.99]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.100]

Given the conditions of 7.5 Comparing, Adding and Subtracting Fractions, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

7.1 Understanding Fractions as Part of a Whole and Collection

Mathematical analysis for Fractions continues on page 176. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Compute $3/8 + 5/12$: $\text{LCM}(8,12) = 24$. $(9 + 10)/24 = 19/24$.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 7.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 7.101]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 7.102]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 7.103]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 7.104]

Given the conditions of 7.1 Understanding Fractions as Part of a Whole and Collection, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

CHAPTER 8

PLAYING WITH CONSTRUCTIONS

CBSE / NCERT Curriculum • Standard Grade 6 Mathematics

CORE TOPICS & LEARNING OBJECTIVES:

- 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares
- 8.2 Constructing Circles of Given Radii
- 8.3 Constructing a Line Segment and its Perpendicular Bisector
- 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors
- 8.5 Drawing Angles using Protractor and Ruler

1. Chapter Introduction & Conceptual Overview

In this chapter, students investigate the foundational principles of Playing with Constructions. Mathematics is not merely about mechanical computations; it is about discovering relationships, finding symmetries, and applying structured reasoning to solve real-world problems.

NCERT Ganita Prakash introduces these concepts through tactile explorations, visual models, and guided inquiries. Students are encouraged to observe patterns carefully before formulating generalized rules.

KEY EXAMPLE 1.1 (WORKED DEMONSTRATION):

Problem Statement: Step-by-step construction of a perpendicular bisector of a 7.2 cm line segment AB.

Step-by-step Solution: Follow standard NCERT methodology with full justifications.

NCERT DIDACTIC EXPLORATION CORNER:

Remember: Always verify your calculation by checking special cases and boundary values.

8.2 Constructing Circles of Given Radii

Mathematical analysis for Playing with Constructions continues on page 178. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw a circle with radius 4.5 cm and mark center O, chord CD and diameter AB.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.2 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.5]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.6]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.7]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.8]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.3 Constructing a Line Segment and its Perpendicular Bisector

Mathematical analysis for Playing with Constructions continues on page 179. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Step-by-step construction of a perpendicular bisector of a 7.2 cm line segment AB.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.3 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.9]

Given the conditions of 8.3 Constructing a Line Segment and its Perpendicular Bisector, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.10]

Given the conditions of 8.3 Constructing a Line Segment and its Perpendicular Bisector, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.11]

Given the conditions of 8.3 Constructing a Line Segment and its Perpendicular Bisector, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.12]

Given the conditions of 8.3 Constructing a Line Segment and its Perpendicular Bisector, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors

Mathematical analysis for Playing with Constructions continues on page 180. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 90° angle using compass and ruler only, explaining each arc.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.13]

Given the conditions of 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.14]

Given the conditions of 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.15]

Given the conditions of 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.16]

Given the conditions of 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.5 Drawing Angles using Protractor and Ruler

Mathematical analysis for Playing with Constructions continues on page 181. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw a circle with radius 4.5 cm and mark center O, chord CD and diameter AB.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.17]

Given the conditions of 8.5 Drawing Angles using Protractor and Ruler, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.18]

Given the conditions of 8.5 Drawing Angles using Protractor and Ruler, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.19]

Given the conditions of 8.5 Drawing Angles using Protractor and Ruler, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.20]

Given the conditions of 8.5 Drawing Angles using Protractor and Ruler, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares

Mathematical analysis for Playing with Constructions continues on page 182. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Step-by-step construction of a perpendicular bisector of a 7.2 cm line segment AB.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.21]

Given the conditions of 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.22]

Given the conditions of 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.23]

Given the conditions of 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.24]

Given the conditions of 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.2 Constructing Circles of Given Radii

Mathematical analysis for Playing with Constructions continues on page 183. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 90° angle using compass and ruler only, explaining each arc.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.25]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.26]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.27]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.28]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.3 Constructing a Line Segment and its Perpendicular Bisector

Mathematical analysis for Playing with Constructions continues on page 184. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw a circle with radius 4.5 cm and mark center O, chord CD and diameter AB.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.29]

Given the conditions of 8.3 Constructing a Line Segment and its Perpendicular Bisector, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.30]

Given the conditions of 8.3 Constructing a Line Segment and its Perpendicular Bisector, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.31]

Given the conditions of 8.3 Constructing a Line Segment and its Perpendicular Bisector, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.32]

Given the conditions of 8.3 Constructing a Line Segment and its Perpendicular Bisector, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors

Mathematical analysis for Playing with Constructions continues on page 185. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Step-by-step construction of a perpendicular bisector of a 7.2 cm line segment AB.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.33]

Given the conditions of 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.34]

Given the conditions of 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.35]

Given the conditions of 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.36]

Given the conditions of 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.5 Drawing Angles using Protractor and Ruler

Mathematical analysis for Playing with Constructions continues on page 186. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 90° angle using compass and ruler only, explaining each arc.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.37]

Given the conditions of 8.5 Drawing Angles using Protractor and Ruler, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.38]

Given the conditions of 8.5 Drawing Angles using Protractor and Ruler, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.39]

Given the conditions of 8.5 Drawing Angles using Protractor and Ruler, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.40]

Given the conditions of 8.5 Drawing Angles using Protractor and Ruler, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares

Mathematical analysis for Playing with Constructions continues on page 187. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw a circle with radius 4.5 cm and mark center O, chord CD and diameter AB.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.41]

Given the conditions of 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.42]

Given the conditions of 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.43]

Given the conditions of 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.44]

Given the conditions of 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.2 Constructing Circles of Given Radii

Mathematical analysis for Playing with Constructions continues on page 188. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Step-by-step construction of a perpendicular bisector of a 7.2 cm line segment AB.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.45]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.46]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.47]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.48]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.3 Constructing a Line Segment and its Perpendicular Bisector

Mathematical analysis for Playing with Constructions continues on page 189. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 90° angle using compass and ruler only, explaining each arc.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.49]

Given the conditions of 8.3 Constructing a Line Segment and its Perpendicular Bisector, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.50]

Given the conditions of 8.3 Constructing a Line Segment and its Perpendicular Bisector, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.51]

Given the conditions of 8.3 Constructing a Line Segment and its Perpendicular Bisector, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.52]

Given the conditions of 8.3 Constructing a Line Segment and its Perpendicular Bisector, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors

Mathematical analysis for Playing with Constructions continues on page 190. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw a circle with radius 4.5 cm and mark center O, chord CD and diameter AB.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.53]

Given the conditions of 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.54]

Given the conditions of 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.55]

Given the conditions of 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.56]

Given the conditions of 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.5 Drawing Angles using Protractor and Ruler

Mathematical analysis for Playing with Constructions continues on page 191. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Step-by-step construction of a perpendicular bisector of a 7.2 cm line segment AB.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.57]

Given the conditions of 8.5 Drawing Angles using Protractor and Ruler, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.58]

Given the conditions of 8.5 Drawing Angles using Protractor and Ruler, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.59]

Given the conditions of 8.5 Drawing Angles using Protractor and Ruler, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.60]

Given the conditions of 8.5 Drawing Angles using Protractor and Ruler, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares

Mathematical analysis for Playing with Constructions continues on page 192. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 90° angle using compass and ruler only, explaining each arc.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.61]

Given the conditions of 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.62]

Given the conditions of 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.63]

Given the conditions of 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.64]

Given the conditions of 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.2 Constructing Circles of Given Radii

Mathematical analysis for Playing with Constructions continues on page 193. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw a circle with radius 4.5 cm and mark center O, chord CD and diameter AB.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.65]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.66]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.67]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.68]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.3 Constructing a Line Segment and its Perpendicular Bisector

Mathematical analysis for Playing with Constructions continues on page 194. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Step-by-step construction of a perpendicular bisector of a 7.2 cm line segment AB.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.69]

Given the conditions of 8.3 Constructing a Line Segment and its Perpendicular Bisector, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.70]

Given the conditions of 8.3 Constructing a Line Segment and its Perpendicular Bisector, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.71]

Given the conditions of 8.3 Constructing a Line Segment and its Perpendicular Bisector, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.72]

Given the conditions of 8.3 Constructing a Line Segment and its Perpendicular Bisector, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors

Mathematical analysis for Playing with Constructions continues on page 195. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 90° angle using compass and ruler only, explaining each arc.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.73]

Given the conditions of 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.74]

Given the conditions of 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.75]

Given the conditions of 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.76]

Given the conditions of 8.4 Constructing Angles of 60°, 90°, 120° and their Bisectors, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.5 Drawing Angles using Protractor and Ruler

Mathematical analysis for Playing with Constructions continues on page 196. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Draw a circle with radius 4.5 cm and mark center O, chord CD and diameter AB.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.77]

Given the conditions of 8.5 Drawing Angles using Protractor and Ruler, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.78]

Given the conditions of 8.5 Drawing Angles using Protractor and Ruler, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.79]

Given the conditions of 8.5 Drawing Angles using Protractor and Ruler, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.80]

Given the conditions of 8.5 Drawing Angles using Protractor and Ruler, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares

Mathematical analysis for Playing with Constructions continues on page 197. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Step-by-step construction of a perpendicular bisector of a 7.2 cm line segment AB.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.81]

Given the conditions of 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.82]

Given the conditions of 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.83]

Given the conditions of 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.84]

Given the conditions of 8.1 The Geometrical Toolkit: Ruler, Compasses, Divider, Set-Squares, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

8.2 Constructing Circles of Given Radii

Mathematical analysis for Playing with Constructions continues on page 198. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Construct a 90° angle using compass and ruler only, explaining each arc.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 8.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 8.85]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 8.86]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 8.87]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 8.88]

Given the conditions of 8.2 Constructing Circles of Given Radii, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

CHAPTER 9

SYMMETRY

CBSE / NCERT Curriculum • Standard Grade 6 Mathematics

CORE TOPICS & LEARNING OBJECTIVES:

- 9.1 Line Symmetry in Natural and Geometric Shapes
- 9.2 Reflection Symmetry and Mirror Lines
- 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle)
- 9.4 Symmetrical Rangoli and Kaleidoscope Art
- 9.5 Applications in Architecture and Nature

1. Chapter Introduction & Conceptual Overview

In this chapter, students investigate the foundational principles of Symmetry. Mathematics is not merely about mechanical computations; it is about discovering relationships, finding symmetries, and applying structured reasoning to solve real-world problems.

NCERT Ganita Prakash introduces these concepts through tactile explorations, visual models, and guided inquiries. Students are encouraged to observe patterns carefully before formulating generalized rules.

KEY EXAMPLE 1.1 (WORKED DEMONSTRATION):

Problem Statement: How many lines of symmetry does a regular hexagon have? (Answer: 6).

Step-by-step Solution: Follow standard NCERT methodology with full justifications.

NCERT DIDACTIC EXPLORATION CORNER:

Remember: Always verify your calculation by checking special cases and boundary values.

9.2 Reflection Symmetry and Mirror Lines

Mathematical analysis for Symmetry continues on page 200. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Identify symmetrical letters in the English alphabet: A, H, I, M, O, T, U, V, W, X, Y.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.2 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.5]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.6]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.7]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.8]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle)

Mathematical analysis for Symmetry continues on page 201. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: How many lines of symmetry does a regular hexagon have? (Answer: 6).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.3 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.9]

Given the conditions of 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.10]

Given the conditions of 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.11]

Given the conditions of 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.12]

Given the conditions of 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.4 Symmetrical Rangoli and Kaleidoscope Art

Mathematical analysis for Symmetry continues on page 202. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Complete the symmetrical figure given a mirror line and half shape.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.13]

Given the conditions of 9.4 Symmetrical Rangoli and Kaleidoscope Art, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.14]

Given the conditions of 9.4 Symmetrical Rangoli and Kaleidoscope Art, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.15]

Given the conditions of 9.4 Symmetrical Rangoli and Kaleidoscope Art, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.16]

Given the conditions of 9.4 Symmetrical Rangoli and Kaleidoscope Art, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.5 Applications in Architecture and Nature

Mathematical analysis for Symmetry continues on page 203. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Identify symmetrical letters in the English alphabet: A, H, I, M, O, T, U, V, W, X, Y.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.17]

Given the conditions of 9.5 Applications in Architecture and Nature, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.18]

Given the conditions of 9.5 Applications in Architecture and Nature, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.19]

Given the conditions of 9.5 Applications in Architecture and Nature, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.20]

Given the conditions of 9.5 Applications in Architecture and Nature, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.1 Line Symmetry in Natural and Geometric Shapes

Mathematical analysis for Symmetry continues on page 204. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: How many lines of symmetry does a regular hexagon have? (Answer: 6).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.21]

Given the conditions of 9.1 Line Symmetry in Natural and Geometric Shapes, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.22]

Given the conditions of 9.1 Line Symmetry in Natural and Geometric Shapes, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.23]

Given the conditions of 9.1 Line Symmetry in Natural and Geometric Shapes, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.24]

Given the conditions of 9.1 Line Symmetry in Natural and Geometric Shapes, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.2 Reflection Symmetry and Mirror Lines

Mathematical analysis for Symmetry continues on page 205. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Complete the symmetrical figure given a mirror line and half shape.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.25]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.26]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.27]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.28]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle)

Mathematical analysis for Symmetry continues on page 206. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Identify symmetrical letters in the English alphabet: A, H, I, M, O, T, U, V, W, X, Y.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.29]

Given the conditions of 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.30]

Given the conditions of 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.31]

Given the conditions of 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.32]

Given the conditions of 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.4 Symmetrical Rangoli and Kaleidoscope Art

Mathematical analysis for Symmetry continues on page 207. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: How many lines of symmetry does a regular hexagon have? (Answer: 6).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.33]

Given the conditions of 9.4 Symmetrical Rangoli and Kaleidoscope Art, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.34]

Given the conditions of 9.4 Symmetrical Rangoli and Kaleidoscope Art, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.35]

Given the conditions of 9.4 Symmetrical Rangoli and Kaleidoscope Art, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.36]

Given the conditions of 9.4 Symmetrical Rangoli and Kaleidoscope Art, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.5 Applications in Architecture and Nature

Mathematical analysis for Symmetry continues on page 208. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Complete the symmetrical figure given a mirror line and half shape.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.37]

Given the conditions of 9.5 Applications in Architecture and Nature, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.38]

Given the conditions of 9.5 Applications in Architecture and Nature, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.39]

Given the conditions of 9.5 Applications in Architecture and Nature, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.40]

Given the conditions of 9.5 Applications in Architecture and Nature, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.1 Line Symmetry in Natural and Geometric Shapes

Mathematical analysis for Symmetry continues on page 209. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Identify symmetrical letters in the English alphabet: A, H, I, M, O, T, U, V, W, X, Y.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.41]

Given the conditions of 9.1 Line Symmetry in Natural and Geometric Shapes, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.42]

Given the conditions of 9.1 Line Symmetry in Natural and Geometric Shapes, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.43]

Given the conditions of 9.1 Line Symmetry in Natural and Geometric Shapes, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.44]

Given the conditions of 9.1 Line Symmetry in Natural and Geometric Shapes, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.2 Reflection Symmetry and Mirror Lines

Mathematical analysis for Symmetry continues on page 210. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: How many lines of symmetry does a regular hexagon have? (Answer: 6).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.45]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.46]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.47]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.48]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle)

Mathematical analysis for Symmetry continues on page 211. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Complete the symmetrical figure given a mirror line and half shape.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.49]

Given the conditions of 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.50]

Given the conditions of 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.51]

Given the conditions of 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.52]

Given the conditions of 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.4 Symmetrical Rangoli and Kaleidoscope Art

Mathematical analysis for Symmetry continues on page 212. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Identify symmetrical letters in the English alphabet: A, H, I, M, O, T, U, V, W, X, Y.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.53]

Given the conditions of 9.4 Symmetrical Rangoli and Kaleidoscope Art, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.54]

Given the conditions of 9.4 Symmetrical Rangoli and Kaleidoscope Art, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.55]

Given the conditions of 9.4 Symmetrical Rangoli and Kaleidoscope Art, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.56]

Given the conditions of 9.4 Symmetrical Rangoli and Kaleidoscope Art, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.5 Applications in Architecture and Nature

Mathematical analysis for Symmetry continues on page 213. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: How many lines of symmetry does a regular hexagon have? (Answer: 6).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.57]

Given the conditions of 9.5 Applications in Architecture and Nature, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.58]

Given the conditions of 9.5 Applications in Architecture and Nature, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.59]

Given the conditions of 9.5 Applications in Architecture and Nature, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.60]

Given the conditions of 9.5 Applications in Architecture and Nature, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.1 Line Symmetry in Natural and Geometric Shapes

Mathematical analysis for Symmetry continues on page 214. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Complete the symmetrical figure given a mirror line and half shape.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.61]

Given the conditions of 9.1 Line Symmetry in Natural and Geometric Shapes, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.62]

Given the conditions of 9.1 Line Symmetry in Natural and Geometric Shapes, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.63]

Given the conditions of 9.1 Line Symmetry in Natural and Geometric Shapes, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.64]

Given the conditions of 9.1 Line Symmetry in Natural and Geometric Shapes, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.2 Reflection Symmetry and Mirror Lines

Mathematical analysis for Symmetry continues on page 215. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Identify symmetrical letters in the English alphabet: A, H, I, M, O, T, U, V, W, X, Y.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.65]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.66]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.67]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.68]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle)

Mathematical analysis for Symmetry continues on page 216. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: How many lines of symmetry does a regular hexagon have? (Answer: 6).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.69]

Given the conditions of 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.70]

Given the conditions of 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.71]

Given the conditions of 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.72]

Given the conditions of 9.3 Figures with Multiple Lines of Symmetry (Equilateral Triangle, Square, Circle), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.4 Symmetrical Rangoli and Kaleidoscope Art

Mathematical analysis for Symmetry continues on page 217. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Complete the symmetrical figure given a mirror line and half shape.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.73]

Given the conditions of 9.4 Symmetrical Rangoli and Kaleidoscope Art, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.74]

Given the conditions of 9.4 Symmetrical Rangoli and Kaleidoscope Art, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.75]

Given the conditions of 9.4 Symmetrical Rangoli and Kaleidoscope Art, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.76]

Given the conditions of 9.4 Symmetrical Rangoli and Kaleidoscope Art, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.5 Applications in Architecture and Nature

Mathematical analysis for Symmetry continues on page 218. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Identify symmetrical letters in the English alphabet: A, H, I, M, O, T, U, V, W, X, Y.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.77]

Given the conditions of 9.5 Applications in Architecture and Nature, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.78]

Given the conditions of 9.5 Applications in Architecture and Nature, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.79]

Given the conditions of 9.5 Applications in Architecture and Nature, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.80]

Given the conditions of 9.5 Applications in Architecture and Nature, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.1 Line Symmetry in Natural and Geometric Shapes

Mathematical analysis for Symmetry continues on page 219. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: How many lines of symmetry does a regular hexagon have? (Answer: 6).

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.81]

Given the conditions of 9.1 Line Symmetry in Natural and Geometric Shapes, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.82]

Given the conditions of 9.1 Line Symmetry in Natural and Geometric Shapes, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.83]

Given the conditions of 9.1 Line Symmetry in Natural and Geometric Shapes, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.84]

Given the conditions of 9.1 Line Symmetry in Natural and Geometric Shapes, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

9.2 Reflection Symmetry and Mirror Lines

Mathematical analysis for Symmetry continues on page 220. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Complete the symmetrical figure given a mirror line and half shape.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 9.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 9.85]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 9.86]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 9.87]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 9.88]

Given the conditions of 9.2 Reflection Symmetry and Mirror Lines, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

CHAPTER 10

RATIO AND PROPORTION

CBSE / NCERT Curriculum • Standard Grade 6 Mathematics

CORE TOPICS & LEARNING OBJECTIVES:

- 10.1 Concept of Ratio as Comparison by Division
- 10.2 Simplifying Ratios to Simplest Form
- 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d)
- 10.4 The Unitary Method for Solving Everyday Problems
- 10.5 Comprehensive Chapter Revision and Practice Assessment

1. Chapter Introduction & Conceptual Overview

In this chapter, students investigate the foundational principles of Ratio and Proportion. Mathematics is not merely about mechanical computations; it is about discovering relationships, finding symmetries, and applying structured reasoning to solve real-world problems.

NCERT Ganita Prakash introduces these concepts through tactile explorations, visual models, and guided inquiries. Students are encouraged to observe patterns carefully before formulating generalized rules.

KEY EXAMPLE 1.1 (WORKED DEMONSTRATION):

Problem Statement: Find the ratio of 40 minutes to 2 hours in simplest form: 40 min / 120 min = 1:3.

Step-by-step Solution: Follow standard NCERT methodology with full justifications.

NCERT DIDACTIC EXPLORATION CORNER:

Remember: Always verify your calculation by checking special cases and boundary values.

10.2 Simplifying Ratios to Simplest Form

Mathematical analysis for Ratio and Proportion continues on page 222. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If the cost of 6 notebooks is Rs 210, find the cost of 14 notebooks: Rs 490.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.2 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.5]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.6]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.7]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.8]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d)

Mathematical analysis for Ratio and Proportion continues on page 223. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the ratio of 40 minutes to 2 hours in simplest form: 40 min / 120 min = 1:3.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.3 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.9]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.10]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.11]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.12]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.4 The Unitary Method for Solving Everyday Problems

Mathematical analysis for Ratio and Proportion continues on page 224. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Check if 15, 45, 40, 120 are in proportion: 15/45 = 1/3 and 40/120 = 1/3. Yes.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.13]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.14]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.15]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.16]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.5 Comprehensive Chapter Revision and Practice Assessment

Mathematical analysis for Ratio and Proportion continues on page 225. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If the cost of 6 notebooks is Rs 210, find the cost of 14 notebooks: Rs 490.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.17]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.18]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.19]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.20]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.1 Concept of Ratio as Comparison by Division

Mathematical analysis for Ratio and Proportion continues on page 226. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the ratio of 40 minutes to 2 hours in simplest form: 40 min / 120 min = 1:3.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.21]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.22]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.23]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.24]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.2 Simplifying Ratios to Simplest Form

Mathematical analysis for Ratio and Proportion continues on page 227. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Check if 15, 45, 40, 120 are in proportion: 15/45 = 1/3 and 40/120 = 1/3. Yes.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.25]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.26]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.27]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.28]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d)

Mathematical analysis for Ratio and Proportion continues on page 228. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If the cost of 6 notebooks is Rs 210, find the cost of 14 notebooks: Rs 490.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.29]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.30]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.31]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.32]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.4 The Unitary Method for Solving Everyday Problems

Mathematical analysis for Ratio and Proportion continues on page 229. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the ratio of 40 minutes to 2 hours in simplest form: 40 min / 120 min = 1:3.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.33]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.34]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.35]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.36]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.5 Comprehensive Chapter Revision and Practice Assessment

Mathematical analysis for Ratio and Proportion continues on page 230. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Check if 15, 45, 40, 120 are in proportion: 15/45 = 1/3 and 40/120 = 1/3. Yes.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.37]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.38]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.39]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.40]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.1 Concept of Ratio as Comparison by Division

Mathematical analysis for Ratio and Proportion continues on page 231. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If the cost of 6 notebooks is Rs 210, find the cost of 14 notebooks: Rs 490.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.41]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.42]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.43]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.44]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.2 Simplifying Ratios to Simplest Form

Mathematical analysis for Ratio and Proportion continues on page 232. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the ratio of 40 minutes to 2 hours in simplest form: 40 min / 120 min = 1:3.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.45]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.46]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.47]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.48]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d)

Mathematical analysis for Ratio and Proportion continues on page 233. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Check if 15, 45, 40, 120 are in proportion: $15/45 = 1/3$ and $40/120 = 1/3$. Yes.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.49]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.50]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.51]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.52]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.4 The Unitary Method for Solving Everyday Problems

Mathematical analysis for Ratio and Proportion continues on page 234. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If the cost of 6 notebooks is Rs 210, find the cost of 14 notebooks: Rs 490.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.53]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.54]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.55]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.56]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.5 Comprehensive Chapter Revision and Practice Assessment

Mathematical analysis for Ratio and Proportion continues on page 235. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the ratio of 40 minutes to 2 hours in simplest form: 40 min / 120 min = 1:3.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.57]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.58]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.59]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.60]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.1 Concept of Ratio as Comparison by Division

Mathematical analysis for Ratio and Proportion continues on page 236. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Check if 15, 45, 40, 120 are in proportion: 15/45 = 1/3 and 40/120 = 1/3. Yes.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.61]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.62]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.63]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.64]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.2 Simplifying Ratios to Simplest Form

Mathematical analysis for Ratio and Proportion continues on page 237. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If the cost of 6 notebooks is Rs 210, find the cost of 14 notebooks: Rs 490.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.65]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.66]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.67]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.68]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d)

Mathematical analysis for Ratio and Proportion continues on page 238. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the ratio of 40 minutes to 2 hours in simplest form: 40 min / 120 min = 1:3.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.69]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.70]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.71]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.72]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.4 The Unitary Method for Solving Everyday Problems

Mathematical analysis for Ratio and Proportion continues on page 239. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Check if 15, 45, 40, 120 are in proportion: 15/45 = 1/3 and 40/120 = 1/3. Yes.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.73]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.74]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.75]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.76]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.5 Comprehensive Chapter Revision and Practice Assessment

Mathematical analysis for Ratio and Proportion continues on page 240. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If the cost of 6 notebooks is Rs 210, find the cost of 14 notebooks: Rs 490.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.77]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.78]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.79]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.80]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.1 Concept of Ratio as Comparison by Division

Mathematical analysis for Ratio and Proportion continues on page 241. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the ratio of 40 minutes to 2 hours in simplest form: 40 min / 120 min = 1:3.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.81]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.82]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.83]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.84]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.2 Simplifying Ratios to Simplest Form

Mathematical analysis for Ratio and Proportion continues on page 242. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Check if 15, 45, 40, 120 are in proportion: 15/45 = 1/3 and 40/120 = 1/3. Yes.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.85]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.86]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.87]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.88]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d)

Mathematical analysis for Ratio and Proportion continues on page 243. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If the cost of 6 notebooks is Rs 210, find the cost of 14 notebooks: Rs 490.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.89]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.90]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.91]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.92]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.4 The Unitary Method for Solving Everyday Problems

Mathematical analysis for Ratio and Proportion continues on page 244. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the ratio of 40 minutes to 2 hours in simplest form: 40 min / 120 min = 1:3.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.93]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.94]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.95]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.96]

Given the conditions of 10.4 The Unitary Method for Solving Everyday Problems, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.5 Comprehensive Chapter Revision and Practice Assessment

Mathematical analysis for Ratio and Proportion continues on page 245. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Check if 15, 45, 40, 120 are in proportion: 15/45 = 1/3 and 40/120 = 1/3. Yes.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.97]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.98]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.99]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.100]

Given the conditions of 10.5 Comprehensive Chapter Revision and Practice Assessment, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.1 Concept of Ratio as Comparison by Division

Mathematical analysis for Ratio and Proportion continues on page 246. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: If the cost of 6 notebooks is Rs 210, find the cost of 14 notebooks: Rs 490.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.101]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.102]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.103]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.104]

Given the conditions of 10.1 Concept of Ratio as Comparison by Division, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.2 Simplifying Ratios to Simplest Form

Mathematical analysis for Ratio and Proportion continues on page 247. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Find the ratio of 40 minutes to 2 hours in simplest form: 40 min / 120 min = 1:3.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.105]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.106]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.107]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.108]

Given the conditions of 10.2 Simplifying Ratios to Simplest Form, evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION:

10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d)

Mathematical analysis for Ratio and Proportion continues on page 248. Students learn to formulate rigorous definitions, evaluate numerical patterns, and apply geometric representations.

FORMULA & CONCEPTUAL FOCUS BOX:

Theorem / Problem: Check if 15, 45, 40, 120 are in proportion: $15/45 = 1/3$ and $40/120 = 1/3$. Yes.

Key Rule: Maintain precision in mathematical notation and verify units at every step.

EXERCISE 10.4 - PRACTICE QUESTIONS

Q1. [NCERT Problem 10.109]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 1. Provide complete working with diagram.

Q2. [NCERT Problem 10.110]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 2. Provide complete working with diagram.

Q3. [NCERT Problem 10.111]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 3. Provide complete working with diagram.

Q4. [NCERT Problem 10.112]

Given the conditions of 10.3 Understanding Proportion: Equality of Two Ratios (a:b :: c:d), evaluate the mathematical statement and determine the value for step 4. Provide complete working with diagram.

NCERT WORKED CANVAS & DIAGRAMMATIC REPRESENTATION: